

Guarded developer toolchain and locked setup

LOCAL EVIDENCE / REMOTE CI NOT YET RUN

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Parent commit: 092781e5a35c5259ceb8ac919f435be799476070

Candidate SHA-256: 46850a7e6b6847c6be434356cf8e44c1762ac988cdd4e772c72acede8f4a4d04

Fingerprint excludes only this report's JSON/TEX/PDF; it is not the final commit SHA.

WHY / problem and value

Give contributors reproducible development inputs and reject setup mismatches before installation, while retaining the existing application and deployment interfaces.

WHAT / change scope

Declare developer Node 24.21.0 and bundled npm 11.19.0. Add explicit doctor/install commands, seven focused tests and setup documentation. Installation delegates to npm ci.

HOW / implementation rationale

Validate an isolated staged snapshot with the declared runtime. Replay both adapter builds, exercise real lock/version rejection, and check three built Node HTTP pages. All 491 locked Node engine constraints accept this version.

WHERE / affected paths

configs/development/toolchain.json
scripts/development/bootstrap.mjs
tests/development/bootstrap.test.mjs
docs/engineering/development.md
CONTRIBUTING.md; docs/README.md

PRESERVED / existing contracts

Application source, public assets, package manifest/lock, adapters, container startup, chart interfaces and CI remain byte-identical to the original application baseline. No automatic platform runtime override.

RISK / what could go wrong

An exact developer pin needs deliberate maintenance. macOS build success does not prove every platform or native dependency path. Install scripts and live-service build inputs still need review.

LIMITATIONS / what is not proved

macOS arm64 only. No Linux, Windows, remote CI or deployment claim. npm warned about seven packages needing install-script approval. Inherited audit advisories remain. Report toolchain unchanged.

ROLLBACK / restore and verify

Revert the development declaration, bootstrap, tests and related guidance in a new reported commit. Existing npm entry points and deployment settings retain their original behavior.

VALIDATION / actual local evidence

Gate	Command / check		Outcome	Duration	Evidence
Locked setup		node scripts/development/bootstrap.mjs install	PASS	11.38 s	artifacts/admission/development/locked-install.log
Bootstrap tests		node -test tests/development/bootstrap.test.mjs	PASS	1.67 s	artifacts/admission/development/bootstrap-tests.log
Node build		npm run build	PASS	253.87 s	artifacts/admission/development/node-build.log
Netlify compile	com-	npm run astro -- build	PASS	302.38 s	artifacts/admission/development/netlify-build.log
Setup rejection	rejec-	python3 -B artifacts/admission/development/check_rejections.py	PASS	0.96 s	artifacts/admission/development/rejection-check.json
Docs source	and	python3 -B artifacts/admission/architecture/check_documentation.py artifacts/admission/development/candidate	PASS	4.74 s	artifacts/admission/development/documentation-check.json

Remote validation at authoring: NOT_RUN. Later CI belongs to the immutable commit's check records, not a follow-up report commit.